

Ujjwal Jain

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EDUCATION

Birla Institute of Technology and Science (BITS) Pilani

Pilani, India

Bachelor of Science in Computer Science; CGPA: 9.3/10

Aug. 2024 – Aug. 2028

OPEN SOURCE CONTRIBUTIONS

Appwrite

Remote

Open Source Contributor

Nov. 2025

- Resolved critical **MFA recovery code validation bug** in production authentication flow affecting user login experience (PR #10925)
- Implemented fix with **comprehensive E2E test coverage** ensuring **consistent CI/CD pass rate**; **merged and deployed to production**

Vitest

Remote

Open Source Contributor

Dec. 2025

- Fixed test execution bug where `test.only` nested in `describe.only` ran unintended tests, preventing CI false passes (PR #9213)
- Refactored test collection logic and added TypeScript diagnostic improvements; **changes merged** after maintainer review
- Currently contributing an in-review fix to Vitest's typechecker improving error handling when `tsc` outputs help text (PR #9214)

PROJECTS

CampusSync | *Next.js, TypeScript, PostgreSQL, Supabase, OCR*

Sep. 2025 – Oct. 2025

- Deployed Multi-Organization SaaS Platform for AI-Powered Certificate Verification and Campus Recruitment with **90+ REST APIs** and **50+ Row-Level Security policies** ensuring complete data isolation
- Optimized database to **sub-100ms** queries using **20+ strategic indexes**; integrated Ed25519 signatures
- Built dual OCR pipeline (Tesseract + Gemini AI) achieving **95%+ accuracy**, reducing verification from minutes to seconds

Fuze | *Flask, React, PostgreSQL, pgvector, Redis*

Aug. 2025 – Present

- Developed an intelligent content management platform that transforms scattered bookmarks into an AI-powered knowledge base with **multi-engine recommendation orchestration** and semantic search using pgvector; optimized model loading by **98%** (**6.7s** → **0.1s**)
- Implemented Redis caching achieving **70-80% hit rate** and **sub-100ms API responses**
- Deployed with **24+ database indexes**, encrypted API key management, and production infrastructure on Hugging Face Spaces

MigrateDB | *TypeScript, Node.js, PostgreSQL, MySQL, SQLite*

Oct. 2025 – Nov. 2025

- Published an NPM package with **100+ installs**: a TypeScript-first database migration tool supporting PostgreSQL, MySQL, and SQLite, featuring safe rollbacks, **100% TypeScript coverage**, and **zero runtime dependencies**
- Implemented Kahn's algorithm (**$O(V+E)$**) for dependency resolution; built CLI with **5 commands** and SHA-256 verification
- Engineered database-level locking using SQL constraints preventing race conditions in distributed CI/CD environments

TECHNICAL SKILLS

Competitive Programming : Codeforces 1150

Languages: Python, TypeScript, JavaScript, Java, SQL, HTML/CSS

Frameworks: Flask, FastAPI, Node.js, Next.js, React, Express.js, Tailwind CSS

Databases: PostgreSQL, MySQL, MongoDB, SQLite, Redis, Supabase, pgvector

DevOps & Tools: Git, GitHub Actions, Docker, CI/CD, npm, Linux

Testing: Vitest, Jest, Playwright, pytest

Core Competencies: Data Structures & Algorithms, Database Systems, Operating Systems, Computer Networks, RESTful API design